Geodesics using Weight Functions

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The Polar Geodesic Equation

Let $U \subseteq \mathbb{C}$ be an open region. In the complex coordinate z, a conformal metric has the form $g = u(z)|\mathrm{d}z|^2$ where u is called a *weight function*. Here $|\mathrm{d}z|^2$ represents the standard complex dot product. At a point $p \in U$ and a vector w starting at p, the norm of w with respect to g is $|w|_g = \sqrt{u(p)}|w|$, where $|w|$ is the standard norm. Note that we've switched from the earlier convention $g = u^2|\mathrm{d}z|^2$ to match the M.R.S. paper.

The connection induced by g is given by $\nabla = \mathrm{d} + \partial \log u$. Evaluating this connection on a function f gives

$$\nabla f = \mathrm{d}f + (\partial_z \log u) f \, \mathrm{d}z \tag{1}$$

In the case where $\gamma : [0, T) \to U$ is a geodesic, we can extend the derivative $\dot{\gamma}$ to a vector field on a tubular neighborhood of $\operatorname{im} \gamma$. Since the tangent bundle of U is trivial, a vector field is the same as a complex function. Thus

$$0 = \nabla_{\dot{\gamma}} \dot{\gamma} = \ddot{\gamma} + (\partial_z \log u) \dot{\gamma} \cdot \langle \mathrm{d}z, \dot{\gamma} \rangle = \ddot{\gamma} + (\partial_z \log u) \dot{\gamma}^2, \tag{2}$$

where $\langle \mathrm{d}z, \dot{\gamma} \rangle$ is the duality pairing[1]. Our weight function will be given in terms of polar coordinates (r, θ). Recall

$$\partial_z = \frac{e^{-i\theta}}{2r}(r\partial_r - i\partial_\theta)$$

so (2) becomes

$$0 = \nabla_{\dot{\gamma}} \dot{\gamma} = \ddot{\gamma} + \frac{e^{-i\theta}}{2}(\partial_r \log u - ir\partial_\theta \log u)\dot{\gamma}^2. \tag{3}$$

We still consider $\dot{\gamma}$ and $\ddot{\gamma}$ to be complex numbers.

[1] In $\mathbb{C}^n$ with coordinates $(z_1, \dots, z_n)$, for a vector $\vec{v} = (v_1, \dots, v_n)$ the pairing is $\langle \mathrm{d}z_i, \vec{v} \rangle = v_i$. In a single dimension where $\vec{v}$ is just a complex number, it's simply $\langle \mathrm{d}z, v \rangle = v$.

Approximating Geodesics on a Cone

Let p and v be the initial position and velocity for the geodesic, both given as complex numbers. Denote

$$\nabla_z = \frac{e^{-i\theta}}{2}(\partial_r \log u - ir\partial_\theta \log u). \tag{4}$$

We will apply Euler's method to

$$\partial_t \dot{\gamma} = -\nabla_z \dot{\gamma}^2. \tag{5}$$

The only difference from our geodesics on the cone is that now we must approximate ∇_z using the finite element data we simulated. To do this, we need to:

- Find the point nearest to $\gamma(t)$, given in polar coordinates;
- Use first differences of $\log u$ to approximate ∇_z in (4);
- Update γ by adding $(\Delta t)\dot{\gamma}$ (here Δt is the step size, not the laplacian);
- Update $\ddot{\gamma}$ according to (5).